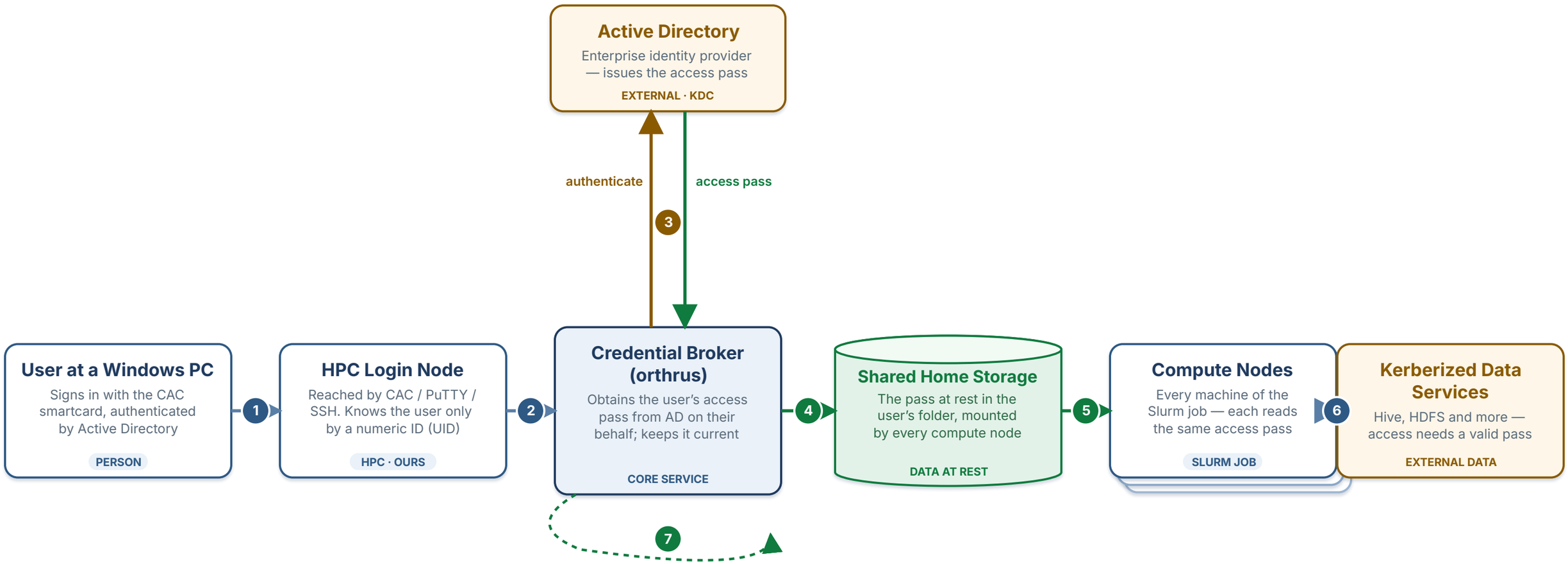
Secure Data Access for HPC Jobs — Executive Data Flow

How a user's identity becomes a short-lived access pass that follows their job to every machine and every data service



How it works — the steps

1. User signs in to the login node with their CAC (as they do today).
2. The login node asks the Credential Broker for that user's pass.
3. The Broker authenticates to Active Directory and receives the pass.
4. The Broker writes the pass into the user's home folder, as the user.
5. Because home is shared, every compute node already sees the pass.
6. Each task shows the pass to the data services and reads its data.
7. The Broker keeps the pass fresh for as long as the job runs.

What flows across every arrow

One thing moves through the system: the user's **access pass** — a short-lived Kerberos ticket that proves who they are. It is never a password, is created only after authentication, and expires on its own.

Person / external data · HPC service (ours) · Data at rest · Access-pass flow

Takeaway: users keep using their CAC; each job automatically gets a temporary, self-expiring pass — no stored passwords, no manual steps.